

Hybridisation and pi-system reasoning — Issue #35 independent candidate: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: From Lewis contributors to one delocalised pi description

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Formal contributors to delocalised orbital model

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Nitrate, NO_3^- , is planar and has three equivalent N–O bonds. Its three conventional Lewis contributors place one

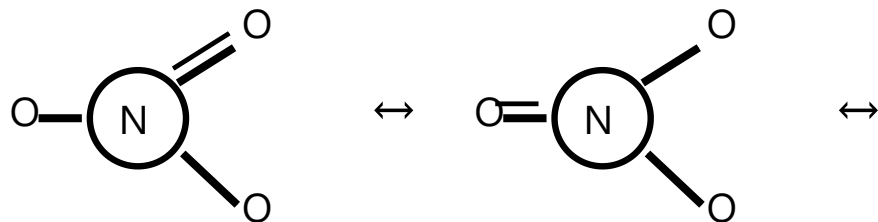
Representation

N=O bond in different positions. Choose between a permanently localised single/double-bond picture and a delocalised pi description. Explain which observations discriminate between them, and state separately the introductory sigma-framework hybridisation model at nitrogen.

Common wrong route · open if you want a warning

+

1 · Compare the contributors



Three Lewis contributors

Need a hint? · guided support

+

Need the concept again?

From Lewis contributors to one delocalised pi description

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: From Lewis contributors to one delocalised pi description

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Formal contributors to delocalised orbital model

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

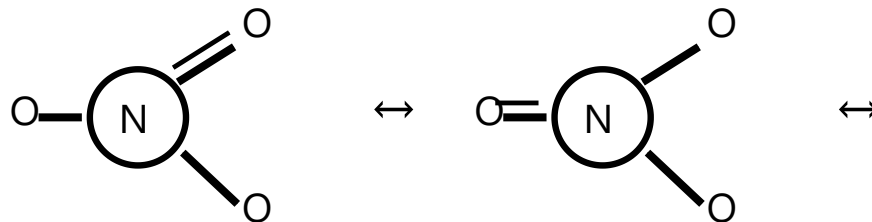
Use the supplied nitrate description: three planar sigma bond directions around nitrogen and a pi system delocalised across the N–O framework. Translate the three Lewis contributors into one qualitative orbital diagram. Distinguish the sigma framework from the delocalised pi system, and explain what is preserved and what must not be interpreted as a molecule switching between drawings.

Common wrong route • open if you want a warning

+

Representation

1 • Compare the contributors



Need a hint? • guided support

+

Need the concept again?

From Lewis contributors to one delocalised pi description

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For propadiene (allene), $\text{H}_2\text{C}=\text{C}=\text{CH}_2$, apply local electron-domain counting at each of the three carbon atoms. Assign the introductory hybridisation of each carbon and show how the two mutually perpendicular unhybridised p orbitals on the central carbon constrain the relative planes of the terminal CH_2 groups.

Representation

Allene connectivity: $\text{H}_2\text{C}=\text{C}=\text{CH}_2$

Construct directions; do not infer from p

Connectivity

Need a hint? · guided support

+

Common wrong route · open if you want a warning

+

Need the concept again?

Coordinate p-orbital directions across several centres

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Ethene is planar; propadiene has terminal CH₂ groups in perpendicular planes. Build a connected account from each carbon's local sigma framework, the required pi overlaps and the p-orbital directions.

Representation

Allene connectivity: H₂C=C=CH₂

Construct directions; do not infer from p

Connectivity

Explain why copying the planar ethene sketch to allene fails.

Need a hint? · guided support

+

Common wrong route · open if you want a warning

+

Need the concept again?

Coordinate p-orbital directions across several centres

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: A reliable atom-local hybridisation procedure

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Atom-local hybridisation procedure

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For the benchmark, use these supplied simplified observations, not an exam-

Representation

source claim: CO₂ is linear with two equivalent C–O bonds; NO₃[–] is planar with three equivalent N–O bonds; NH₃ is pyramidal with three equivalent N–H bonds. Assess the claim 'Equivalent bonds imply sp² at the central atom.' Use counterexamples and valid warrants, and distinguish equivalence, electron-domain count and hybridisation as a model description.

Common wrong route · open if you want a warning

+

1 · One at a time

Choose one atom

Need a hint? · guided support

+

Need the concept again?

A reliable atom-local hybridisation procedure

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: A reliable atom-local hybridisation procedure

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Atom-local hybridisation procedure

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For propyne, $\text{CH}_3\text{--C}\equiv\text{CH}$, assign the introductory hybridisation of each carbon and count the sigma and pi bonds in the entire molecule. Show a check that distinguishes the triple bond's components from the local domain count.

Common wrong route · open if you want a warning

+

Representation

1 • One at a
time

Choose one atom

Need a hint? • guided support

+

Need the concept again?

A reliable atom-local hybridisation procedure

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: From Lewis contributors to one delocalised pi description

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Formal contributors to delocalised orbital model

Question type: Constructed response

Source: Owner-supplied question

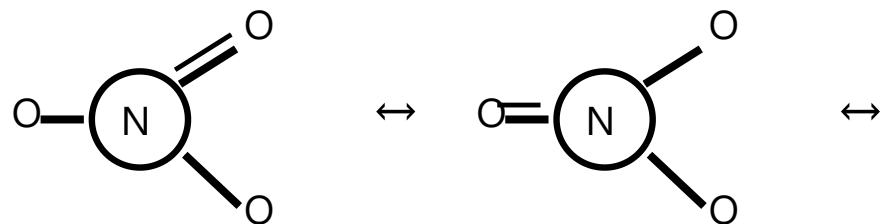
Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies that carbonate, CO_3^{2-} , is planar and has three equivalent C–O bonds; its three conventional Lewis contributors place one C=O bond in different positions. Construct one qualitative description of its sigma framework and delocalised pi system. State why no single contributor should be read as the complete physical picture.

Representation

1 · Compare the contributors



Three Lewis contributors

Common wrong route · open if you want a warning

+

Need a hint? · guided support

+

Need the concept again?

From Lewis contributors to one delocalised pi description

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Compare rotation about the central C–C bond in ethane with twisting about the C=C bond in ethene. Use the sigma and pi overlap arrangements to explain which overlap is altered by each motion. Do not

Representation

1 · Sigma axis



infer a numerical energy barrier from a sketch.

Common wrong route · open if you want a warning

+

Sigma axis

Need a hint? · guided support

+

Need the concept again?

Coordinate p-orbital directions across several centres

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ in an approximately planar conformation, identify the

Representation

continuous sequence of p orbitals and contrast it with $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$. Explain which structural feature interrupts that pathway in the second molecule.

Common wrong route · open if you want a warning

+

1 · Sigma axis



Sigma axis

Need a hint? · guided support

+

Need the concept again?

Coordinate p-orbital directions across several centres

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: A reliable atom-local hybridisation procedure

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Atom-local hybridisation procedure

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student assigns one hybridisation label to an entire molecule. Use propyne and allene to test this procedure and devise a more reliable local procedure. State the electron-domain information and orbital-consistency checks your procedure needs.

Common wrong route · open if you want a warning

+

Representation

1 · One at a
at a time

Choose one atom

Need a hint? · guided support

+

Need the concept again?

A reliable atom-local hybridisation procedure

Chemistry · Hybridisation and pi-system reasoning — Issue #35 independent candidate